

# Pratik Dhulubulu

✉ pratikdhulubulupsd3@gmail.com 📞 +91 7057416720 in Pratik Dhulubulu 🌐 pratikdhulubulu

## Skills

**Programming Languages:** Embedded C, C++, Python

**Embedded Systems:** 16-bit, 32-bit Microcontroller and Microprocessors, RTOS, Communication Protocols, Drivers, Data Structures & Algorithms, Linux, Debugging, Version Control & Build Systems.

**Tools & Platforms:** Visual Studio Code, Git, GitHub Copilot, GDB, Make, PC Lint, Wireshark, Flashing & Debugging Tools, Qt Creator, FreeRTOS, VectorCAST, Vector CAN Devices, Jira, Serial Terminals, Keil µVision, Eclipse, STM32CubeIDE.

## Experience

**Embedded Software Engineer | Knorr-Bremse TCI, Pune**

July 2026 – Present

- Development & maintenance of embedded software modules for the EP2002 Classic Rail Braking System & CCMT Datalogger.
- Involved in analysis and resolution of Jira issues in system software across three maintenance releases.
- Code reviews, Module Specification design reviews, and automated generation of Module Specification documents.
- Leading a team of 3 engineers for module testing activities across multiple maintenance releases.
- Review of unit test reports and test specifications, along with maintenance of code deviation reports.
- Setup and Debugging of test rigs for SAT, Regression test, CAN, MVB & TRDP protocols maintenance and testing.

**Embedded Software Trainee Engineer | Knorr-Bremse TCI, Pune**

July 2025 – June 2026

- Supported development of Ethernet Maintenance Link and Utility modules for the EP2002 Classic CCMT Datalogger system.
- Supported software debugging and bug fixing activities for maintenance releases of the EP2002 Classic Rail Braking System.
- Implemented test automation using VectorCAST Project for build environments, test case execution, and report generation.
- Automated generation of test specification documents using GitHub Copilot, improving efficiency & consistency of documents.

**Embedded Software Intern | Knorr-Bremse TCI, Pune**

Aug 2024 – July 2025

- Performed unit testing of electro-pneumatic break system software modules and drivers using VectorCAST tool.
- Development and execution of test cases. Identified and reported software defects and updated Test Reports.

**Embedded Software Development Intern | Emerson, Pune**

Jan 2024 – July 2024

- Supported HMI development using Qt Creator for Ultrasonic Plastic Welding Platform for Industrial Automation.
- Developed C++ backend & model based UI with QML for HMI system controlling various Supervisory Controllers.

## Education

**D.K.T.E. Textile and Engineering Institute, Kolhapur**

Jan 2021 – June 2024

B.Tech Electronics & Telecommunication Engineering | CGPA: 8.04

## Projects

**EP2002 Classic Rail Braking System**

- Distributed electro-pneumatic braking system used in metro and rail vehicles to control and monitor electro-pneumatic brake valves in real time, ensuring reliable and deterministic braking under strict timing and safety constraints.
- Development of system software, issue resolution across maintenance releases, design specification update & reviews.
- Module test reviews, setting up system test rig for CAN, MVB & TRDP communication debugging & tests.

**CCMT Data-Logger - Central Control & Monitoring Terminal**

- Critical diagnostic and monitoring device within the rail braking system. It is responsible for supervising brake system behavior, collecting operational data, and supporting fault analysis and maintenance activities.
- Supported development of Ethernet Maintenance Link and Utility modules and integration with brake system.

**Polaris - Ultrasonic Plastic Welding Platform**

- An advanced & flexible plastic joining platform with an intuitive and user-friendly HMI control designed to support a constant demand for the assembly of smaller and more intricate plastic components.
- Supported Front-End development for designing responsive and intuitive UI according to Wire-frames using QML.
- Developed Back-End in C++ for data acquisition and communication between HMI and various Supervisory Controllers.

**Smart Glove Controlled Robotic Arm**

- Developed a Wearable Smart Glove that recognizes hand gestures and controls the Robotic Arm operations wirelessly.
- Developed C++ firmware for sensor data, gesture mapping and Arm control, also involved in testing and sensor calibration.

## Achievements

- **Competitive Programming Leetcode** : Global rank of 3163 and a highest rating of 1584 in LeetCode contest 418.